



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL AIR DEFENCE

MADE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

## SOVEREIGN UNMANNED AERIAL SYSTEMS PROGRAM

# SOVEREIGN AUTONOMOUS SWARM & AIR DEFENCE UAV CAPABILITY

A dedicated sovereign unmanned aerial warfare architecture engineered to Indian Armed Forces requirements—delivering autonomous multi-tier swarm reconnaissance, attrition-tolerant mesh command & control, and high-altitude forward border overwatch.

## PILLAR 01 TACTICAL SWARM &amp; ISR



## SWARM & OVERWATCH ISTAR

[ SECT 01 ]

## MULTI-LAYER DISTRIBUTED SURVEILLANCE &amp; MESH C2

Coordinated heterogeneous swarm linking disposable micro Nano-scouts (350–450g) and long-endurance Overwatch UAV relays (~20kg) via self-healing peer-to-peer data-links.

- Decentralized autonomous collision avoidance and patrol logic
- Real-time EO/IR thermal target tracking streamed to tactical GCS
- Attrition-tolerant multi-node coordination in contested airspace

## SWARM NODE SIZE

6 to 24+ Units

## OPERATING CEILING

8,000+ ft AMSL

## OVERWATCH MTOW

~20 kg Class

## C2 DATA-LINK

Secure Mesh RF

## PILLAR 02 CONTESTED C2 &amp; EW HARDENING



## TACTICAL GCS & RESILIENT C2

[ SECT 02 ]

## ANTI-JAM MESH TELEMETRY &amp; SOVEREIGN NAVIC SYNC

Ruggedized tactical ground command console and anti-jamming RF communication architecture engineered for contested electronic warfare environments and GPS-denied forward borders.

- Unified C2 workstation managing simultaneous multi-UAV swarm sorties
- Frequency-hopping spread spectrum with optical dead-reckoning fallback
- SORA SAIL II/III military safety compliance with emergency FTS termination

## C2 DATALINK

P2P Encrypted Mesh

## OPERATIONAL RADIUS

100+ km BVLOS

## NAVIGATION

NAVIC + Anti-Jam INS

## SAFETY CLEARANCE

SORA SAIL II/III

## SOVEREIGN AIR DEFENCE DOCTRINE

UNIFIED INDIAN ARMED FORCES ARCHITECTURE

- Autonomous Swarm Intelligence: Decentralized bounding overwatch and collaborative targeting
- High-Altitude Border Envelope: Sub-zero temperature and Himalayan wind gradient rated
- Contested Spectrum Hardening: NAVIC satellite sync + frequency-hopping RF data-links
- 100% Sovereign IPR: Indigenous carbon-composite fabrication and iDEX / MoD alignment

8,000+ ft AMSL

## HIGH-ALTITUDE OPERATIONAL ENVELOPE

Engineered for forward Himalayan posts, steep valley patrols, and sub-zero border warfare.

50-150 km

## ENCRYPTED SWARM MESH RADIUS

BVLOS frequency-hopping RF telemetry with peer-to-peer relay for EW-contested environments.

6-24+ Nodes

## HETEROGENEOUS SWARM FLEET

Decentralized teaming of sub-500g nano-scouts and 20kg high-altitude overwatch mothership relays.

Make in India

## SOVEREIGN IPR &amp; ASSEMBLY

Complete indigenous production roadmap aligned with iDEX and Ministry of Defence procurement guidelines.